



DALHOUSIE UNIVERSITY

CSCI 5308: Advanced Topics in Software Development

Bonus Assignment

Banner ID: B01026328

Name: Parva Mineshbhai Patel

Date: 2025 – 04 – 02

Pattern Name: Observer Pattern

Category: Behavioral Pattern

Location:

src/main/java/com/dalhousie/dalhousie_marketplace_backend/service/MessageService.java

Fork Link: <https://github.com/2612-Parva/Group03>

Github Link: <https://github.com/2612-Parva/Group03/commit/8b5e66fbdf1b11b487d33874864e82ff40b99c24>

Explanation: The message service directly handled both message delivery and notifications in a tightly coupled manner. So, this implementation violated the Single Responsibility Principle by mixing message handling with notification logic and also made it difficult to add new notification mechanisms. So, the Observer pattern was implemented in the MessageService to decouple the core message handling logic from notification mechanism. Moreover, by introducing MessageSubject and MessageObserver interfaces with concrete NotificationObserver and WebSocketObserver implementations, I achieved loose coupling between components while preserving all existing functionality. Furthermore, this makes the system more maintainable as new notification types can be added without modifying the core message service.

References

- 1) Observer in Java: Behaviour Pattern , " Refactoring.Guru [Online].
Available: <https://refactoring.guru/design-patterns/observer/java/example>. [Accessed: April 02, 2025].